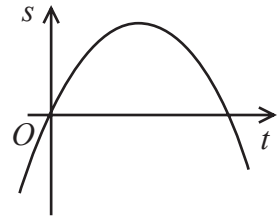
14 A particle P is moving in a straight line.
O is a fixed point on the straight line.
The distance, s metres, of P from O at time t seconds is given by

$$s = 80t - 5t^2$$

Use algebra to find the greatest distance of P from O when $0 \leqslant t \leqslant 16$



$s = -5(t^2 - 16t)$

1 mark
Factor out −5

×	t	-8
t	t^2	$-8t$
-8	$-8t$	64

$t^2 - 16t = (t - 8)^2 - 64.$

$s = -5[(t - 8)^2 - 64] = 320 - 5(t - 8)^2.$

1 mark
Complete the square

$(t - 8)^2 \geq 0$, so $s \leq 320$.

Equality at t=8, which is within $0 \leq t \leq 16$.

1 mark
Substitute t=8

$s(8) = 80 \times 8 - 5 \times 8^2 = 640 - 320 = 320$m.

1 mark
Greatest distance320m

.....metres

(Total for Question 14 is 4 marks)